

Lab 1

Problem Statement 10: Student Club Event Ticketing & Budget Portal

1. Requirements Table

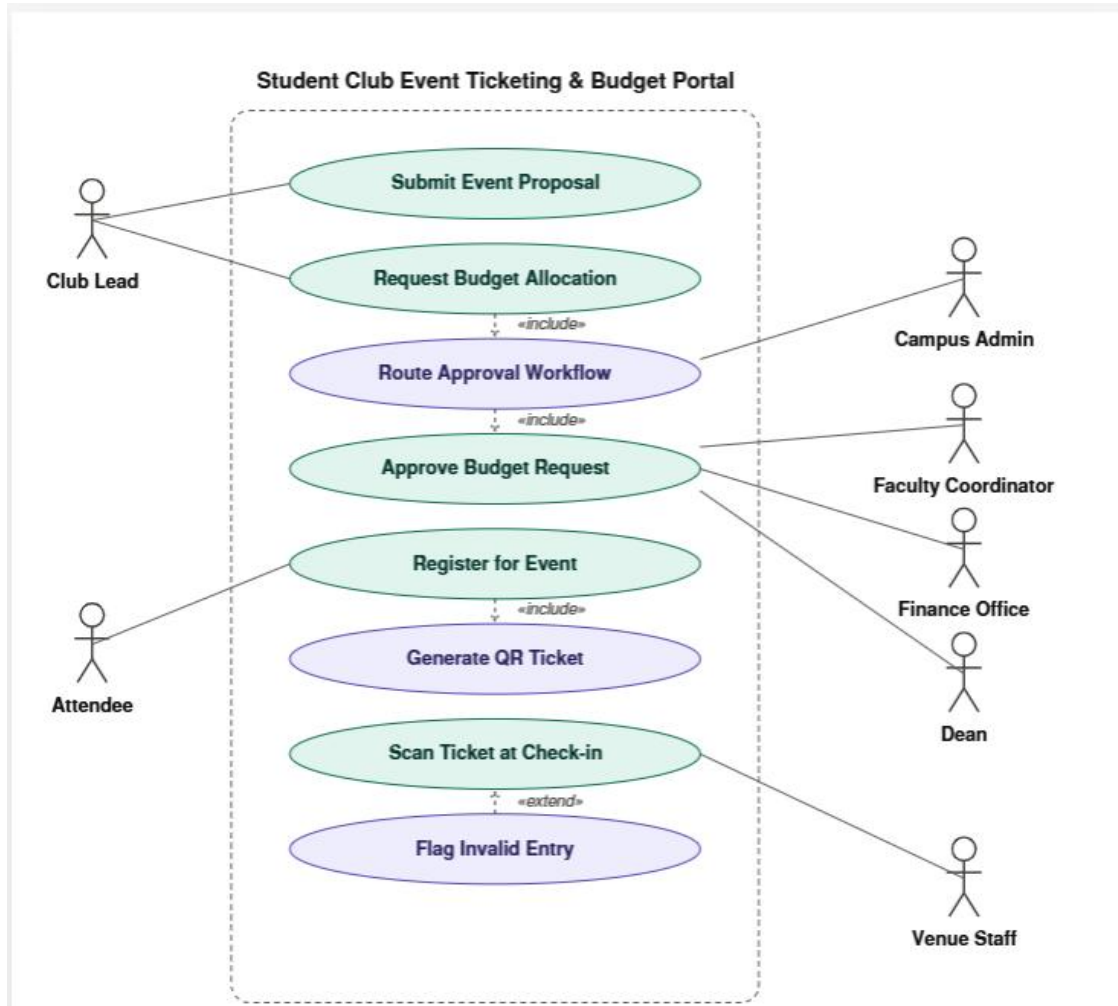
Functional Requirements

Req ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall provide Club Leads with a form through which they can enter event information and specify the amount of university funding they require.	High	Pass: The proposal is stored after all mandatory event and budget details are entered. Fail: Submission is prevented when mandatory details are missing.	Allows clubs to formally request support for their proposed events.
FR-002	The system shall forward a submitted funding request to the Faculty Coordinator, Finance Office, and Dean in the defined approval order.	High	Pass: Each stage receives the request only after the preceding stage has completed its approval. Fail: A request moves ahead while an earlier approval is incomplete.	Maintains the university's required approval sequence.
FR-003	The system shall enable authorized officials to record either an approval or rejection for a submitted event funding request.	High	Pass: The decision, approving official, and current status are saved successfully. Fail: A user without approval privileges is able to change the decision.	Ensures that only responsible officials can make approval decisions.
FR-004	The system shall create an individual QR-based ticket when a participant successfully registers for an approved event.	High	Pass: Every successful registration receives one unique QR ticket. Fail: A registration receives no ticket or receives the same ticket as another registration.	Provides attendees with a digital method of proving their event registration.
FR-005	The system shall enable authorized venue personnel to scan an attendee's QR ticket and update its entry status.	High	Pass: A valid unused ticket is accepted and changed to an attended/used status. Fail: An invalid or previously used ticket is accepted.	Prevents unauthorized or repeated entry at the event venue.

Non-Functional Requirements

Req ID	Description	Priority	Acceptance Criteria	Rationale
NFR-001	The QR-ticket verification service shall complete authenticity checking within 100 milliseconds during venue entry operations.	High	Pass: Performance testing under expected peak usage records a verification time below 100 ms. Fail: Verification takes 100 ms or longer.	Fast verification prevents delays when many attendees arrive together.
NFR-002	The system shall enforce role-based access so that users can view or change only the approval and ticket information permitted for their role.	High	Pass: Access-control testing blocks unauthorized viewing and modification attempts. Fail: A user can access or alter information outside their assigned permissions.	Protects event, approval, and ticket information from unauthorized activity.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: Request Budget Allocation

Primary Actor: Club Lead

Secondary Actors: Faculty Coordinator, Finance Office, Dean (via the included Route Approval Workflow)

Related Use Cases: «include» Route Approval Workflow → «include» Approve Budget Request

Trigger: Club Lead requests funding for an already-submitted event proposal.

Preconditions

- Club Lead is authenticated and logged in to the portal.
- The event has an existing proposal that has been submitted through the Submit Event Proposal use case.
- No budget request is already pending for this event.

Postconditions (success)

- Budget request record exists with status “Approved” and a finalized allocated amount.
- The event proposal is linked to its approved budget.
- Club Lead has received a notification of the outcome.

Postconditions (failure)

- Budget request record exists with status “Rejected”, including a rejection reason and the stage at which it was rejected.

Main Success Scenario

1. Club Lead selects “Request Budget Allocation” for the submitted event proposal.
2. System displays a budget request form pre-filled with event details, and prompts for the requested amount, an itemized expense breakdown, and justification.
3. Club Lead enters the requested amount, itemized budget lines, and justification, then submits the form.
4. System validates that all mandatory fields are complete and the requested amount is within policy limits.
5. System creates the budget request and triggers Route Approval Workflow, setting status to “Pending Faculty Coordinator Review.”
6. Faculty Coordinator reviews the request and digitally signs off; system updates status to “Pending Finance Office Review.”
7. Finance Office reviews the request and digitally signs off; system updates status to “Pending Dean Review.”
8. Dean reviews the request and grants final sign-off; system executes Approve Budget Request, setting status to “Approved” and recording the allocated amount.
9. System notifies Club Lead of the approval and updates the event's available budget balance.
10. Use case ends successfully.

Alternate Flow — 6a. Rejected During Approval Chain

- 6a1. At any approval stage (Faculty Coordinator, Finance Office, or Dean), the reviewer rejects the request and enters a rejection reason.
- 6a2. System sets the budget request status to “Rejected” and halts the workflow; it is not forwarded to the next approver.
- 6a3. System notifies Club Lead of the rejection, including the reason and the stage at which it occurred.
- 6a4. Club Lead may revise the request and resubmit, which restarts the Route Approval Workflow from Faculty Coordinator review.